

BAAZ & UDAY JKBOSE

2026 CLASS 11TH

SOME BASIC CONCEPT OF CHEMISTRY

LECTURE-7

CHEMISTRY

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Topics To Covered



1

PERCENTAGE COMPOSITION

2

EMPIRICAL FORMULA

3

MOLECULAR FORMULA AND ITS DETERMINATION

} Important
Topic

REVISE



Mole concept $\rightarrow$

Numerical formula.

$$1) \text{ No. of mole of atom} = \frac{\text{Given atomic mass}}{\text{Atomic mass}}$$

$$2) \text{ No. of " " molecule} = \frac{\text{Given molecular mass}}{\text{Molecular mass.}}$$

$$3) \text{ No. of moles of Gas} = \frac{\text{Given Volume}}{22.4 \text{ L.}}$$

$$4) \text{ No. of molecules} = \text{No. of moles} \times N_A$$

$$5) \text{ No. of atoms} = \text{No. of molecules} \times \text{Atoms}$$

$$6) \text{ Mass of an atom} = \frac{\text{Atomic mass}}{N_A}$$

$$7) \text{ Mass of molecule} = \frac{\text{Molecular mass}}{N_A}$$

Percentage composition

$$= \frac{\text{Mass of atom in that compound} \times 100}{\text{Total mass of a compound}}$$

EMPIRICAL AND MOLECULAR FORMULA $\Rightarrow$

MOLECULAR FORMULA $\Rightarrow$ It is defined as the formula that represent exact no. of different types of atoms present in a compound.

EMPIRICAL FORMULA $\Rightarrow$ It is defined as the formula that represents simplest whole no. ratio of different types of atoms present in a molecule.

For example. Molecular Formula of benzene is C_6H_6 .
means in one molecule of benzene there are six-C-atoms and six hydrogen atoms.

But its empirical formula is simplest whole no. ratio i.e. $6:6$
 $= 1:1$ i.e. CH.

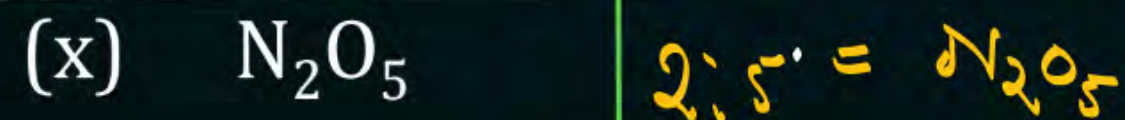
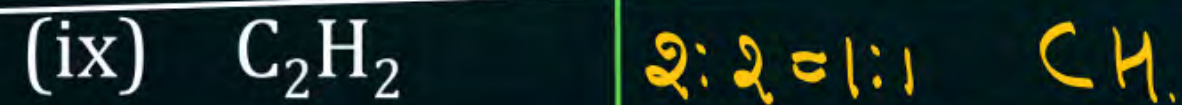
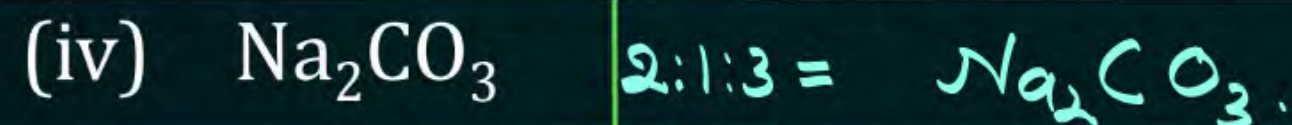
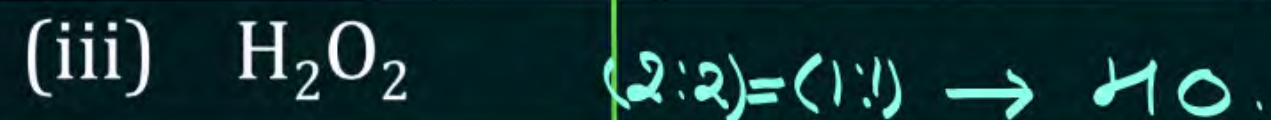
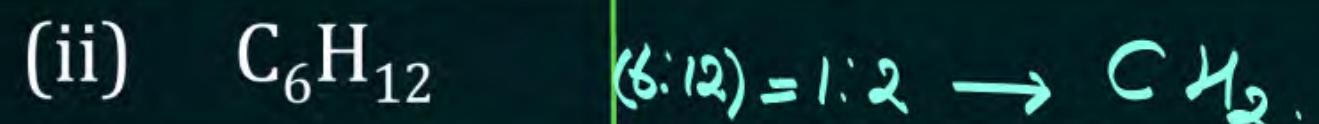
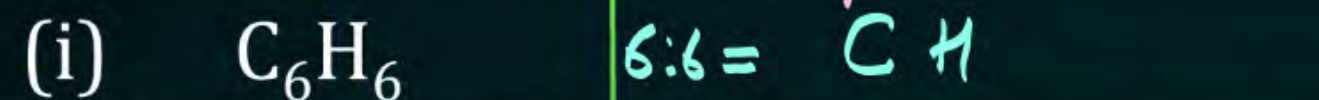
- We can determine the empirical formula of any compound if per centage composition is given & molecular formula can also be calculated if molar mass is also given.

QUESTION



Write the empirical formula of the compounds having the molecular formula:

Molecular Formula Empirical Formula.



NOTE $\rightarrow$ i) Two different molecules can have same empirical formula.

ii) In some cases it is possible to have same Molecular & Empirical formula.

Relation between Empirical formula & Molecular formula.

These two following relationship

$$\text{Molecular formula} = n \times \text{Empirical formula}$$

$$n = 1, 2, 3, 4, 5, \dots \text{ (Whole no.)}$$

'n' can be determined by.

$$n = \frac{\text{Molecular Mass}}{\text{Empirical formula mass}}$$

For example, 'n' for benzene (78 g) can be determined by.

$$n = \frac{\text{Molecular Mass of } C_6H_6}{\text{Empirical formula mass of } C_6H_6}$$

$$= \frac{78}{13} = 6$$

Empirical formula mass

$$= CH$$

$$= 12 + 1$$

$$= 13g$$

QUESTION



How much copper can be obtained from 100 g of copper sulphate (CuSO_4)? *(Given mass of Cu = 63.5g)*

Sol: $\rightarrow$ Percentage composition = $\frac{\text{Mass of element in a compound}}{\text{Total mass of compound}} \times 100$.

$$= \frac{63.5}{63.5 + 32 + (4 \times 16)} \times 100$$

$$= \frac{63.5}{159.5} \times 100 = 39.81\%$$

Mass of an element present in 100g of compound.

Cu

CuSO₄

QUESTION

Home Work for Community



Calculate the mass per cent of different elements in sodium sulphate (Na_2SO_4).

or
percentage composite

i) Mass percent of Na = $\frac{\text{Mass of element in compound}}{\text{Total mass of compound}} \times 100$

$$= \frac{46}{(2 \times 23) + 32 + (4 \times 16)} \times 100 = \frac{46}{46 + 32 + 64} \times 100$$

ii) S

$$= \frac{32}{142} \times 100$$

iii) O

$$= 32.39\%$$

$$\text{Na} = 23$$

$$\text{S} = 32$$

$$\text{O} = 16$$

$$C1 = 35.5$$

$$C1_2 = \underline{35.5} \times$$

$$35.5 \times 2 = \textcircled{71g}$$

QUESTION



A compound contains 4.07% hydrogen, 24.27% carbon and 71.65% chlorine.

Its molar mass is 98.96 g. What are its empirical and molecular formulas?

Sol.

Given mass

★ (Khayal)

Element	Percentage composition (g)	Atomic mass	No. of moles = $\frac{\text{Given atomic mass}}{\text{Atomic mass}}$	Molar Ratio	Simple whole no. Ratio
H	4.07g	1g	$\frac{4.07}{1} = 4.07$	$\frac{4.07}{2.01} = 2$	2
C	24.27g	12g	$\frac{24.27}{12} = 2.02$	$\frac{2.02}{2.01} = 1$	1
Cl	71.65g	35.5	$\frac{71.65}{35.5} = 2.01$	$\frac{2.01}{2.01} = 1$	1

Therefore empirical formula is = CH_2Cl ✓

$$\therefore \text{Molecular formula} = n \times \text{Empirical formula} \\ = n \times \text{CH}_2\text{Cl}$$

$$\therefore n = \frac{\text{Molecular Mass}}{\text{Empirical formula mass}} = \frac{98.96}{49.5} = 2$$

$$\therefore \text{Molecular Formula} = n \times \text{CH}_2\text{Cl} \\ = 2 \times \text{CH}_2\text{Cl} \\ = \text{C}_2\text{H}_4\text{Cl}_2 \quad \checkmark$$

Empirical formula mass of CH_2Cl	$= 12 + 2 + 35.5$
	$= 49.5$

QUESTION



Determine the molecular formula of an oxide of iron, in which the mass per cent of iron and oxygen are 69.9 and 30.1, respectively. (Given: $\rightarrow$ Vapour Density of Oxide = 79.9).

Sol: $\rightarrow$

★

$Fe_{\text{mass}} = 56 \text{ g}$

Element	Given mass	Atomic mass	No. of moles = $\frac{\text{Given mass}}{\text{Atomic mass}}$	Molar Ratio	Simplest Whole no. Ratio.
Fe	69.9	56	$\frac{69.9}{56} = 1.23$	$\frac{1.23}{1.23} = 1$	$1 \times 2 = 2$
O	30.1	16	$\frac{30.1}{16} = 1.88$	$\frac{1.88}{1.23} = 1.5$	$1.5 \times 2 = 3$

Empirical formula = Fe_2O_3 .

Molecular Formula = $n \times$ Empirical formula

$$\Rightarrow n = \frac{\text{Molecular Mass}}{\text{Empirical mass.}}$$

$$= \frac{159.8}{160} \approx \textcircled{1}$$

$$\begin{aligned} \therefore \text{Molecular Formula} &= n \times \text{Empirical formula} \\ &= 1 \times \text{Fe}_2\text{O}_3 \\ &= \text{Fe}_2\text{O}_3. \end{aligned}$$

Relationship between Molecular mass
+ Vapour density.

$$\text{Molecular mass} = 2 \times \text{Vapour density.}$$

$$= 2 \times 79.9 = \underline{159.8}$$

$$\begin{aligned} \text{Fe}_2\text{O}_3 &= 2 \times 56 + 3 \times 16 \\ &= 112 + 48 \\ &= 160 \end{aligned}$$

QUESTION

What is the mass percent of carbon in ethanol ($\text{C}_2\text{H}_5\text{OH}$)?

1 13.13%

2 34.73%

3 51.14%

4 88.79%

QUESTION

Which of the following represents the simplest whole number ratio of elements in a compound?

- 1 Molecular formula
- 2 Empirical formula
- 3 Structural formula
- 4 Ionic formula

QUESTION



Match the following terms with their correct descriptions:

Column – I		Column – II	
(A)	Empirical formula	1.	Shows the simplest ratio of atoms
(B)	Molecular formula	2.	Shows the actual number of atoms
(C)	Percentage composition	3.	Mass of an element per 100g of compound
(D)	Molar mass	4.	Mass of one mole of a substance

a) $A=2, B=3, C=4, D=2$

b) $A=1, B=2, C=4, D=3$

c) $A=1, B=2, C=3, D=4$

d) $A=4, B=3, D=2, C=1$

REVISE KAR K Aana.

$$\begin{array}{l} 1.5 \times 1 = 1.5 \\ \quad \times 2 = \textcircled{3} \end{array}$$

Thank
You